

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458418.627 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.012
Transit depth (Rp/Rs)^2	2.5 +/- 0.39 [%]
Orbital Inclination (inc)	84.16 +/- 1.47
Airmass coefficient 1 (a1)	1.157 +/- 0.01
Airmass coefficient 2 (a2)	-0.0988 +/- 0.0099
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	None
Optimal Aperture	4.58
Optimal Annulus	11.25
Transit Duration (day)	0.064 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.012 vs 0.1326 (deviation 2.12σ)
Duration fit vs published	0.064 d vs 0.0737 d (deviation 0.42σ)
Mid-time O–C	-12.9 min
Depth SNR	13.2
Residual scatter	0.87 %

Light curve

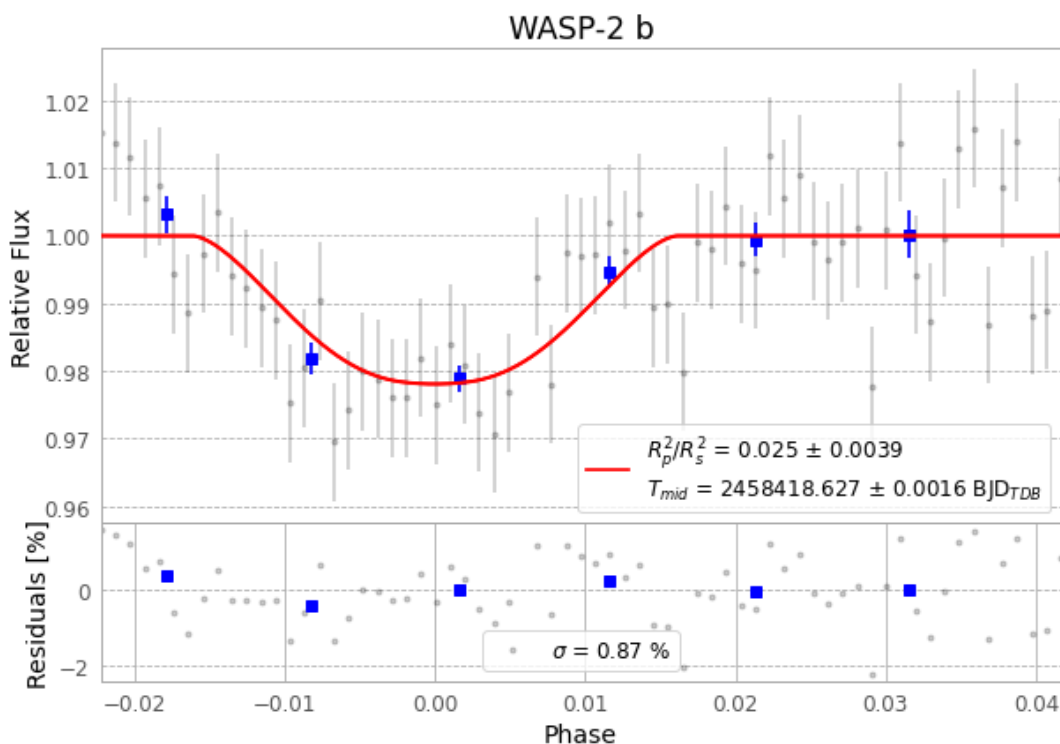


Figure 1 — FinalLightCurve_WASP-2 b_2018-10-26.png (SHA-256 c357dd9b55df465c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [264, 280], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fac9c931ba7185a4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

67 FITS frames (44 MB), WASP-2181027015112.FITS → WASP-2181027050912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-181027.log (5912c8e778c8...), FinalLightCurve_WASP-2 b_2018-10-26.png (c357dd9b55df...), FinalLightCurve_WASP-2 b_2018-10-26.csv (72b445e495c6...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 20:50Z.